Password Complexity Validator

Using Core Java & OOP Principles

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Problem Statement

The Security Challenge

Weak passwords are the #1 entry point for cyber attacks. Users often prioritize ease of memory over security strength.

- easy-to-guess credentials
- lack of complexity enforcement
- increased risk of data theft

Project Objectives

Validate Strength

Automate password analysis based on length, digits, and character casing.

Apply OOP

Implement core Java principles like Encapsulation to secure data flow.

Modular Logic

Separate account data from validation logic for cleaner maintenance.

Technology Stack

Software Requirements

- Java JDK 17+
- VS Code / Eclipse IDE
- Windows Platform

Core Concepts Applied

- ✓ Conditional Statements (if-else)
- ✓ String Methods (length, charAt)
- ✓ OOP: Classes & Encapsulation

Working Methodology

Input
User enters name & pass

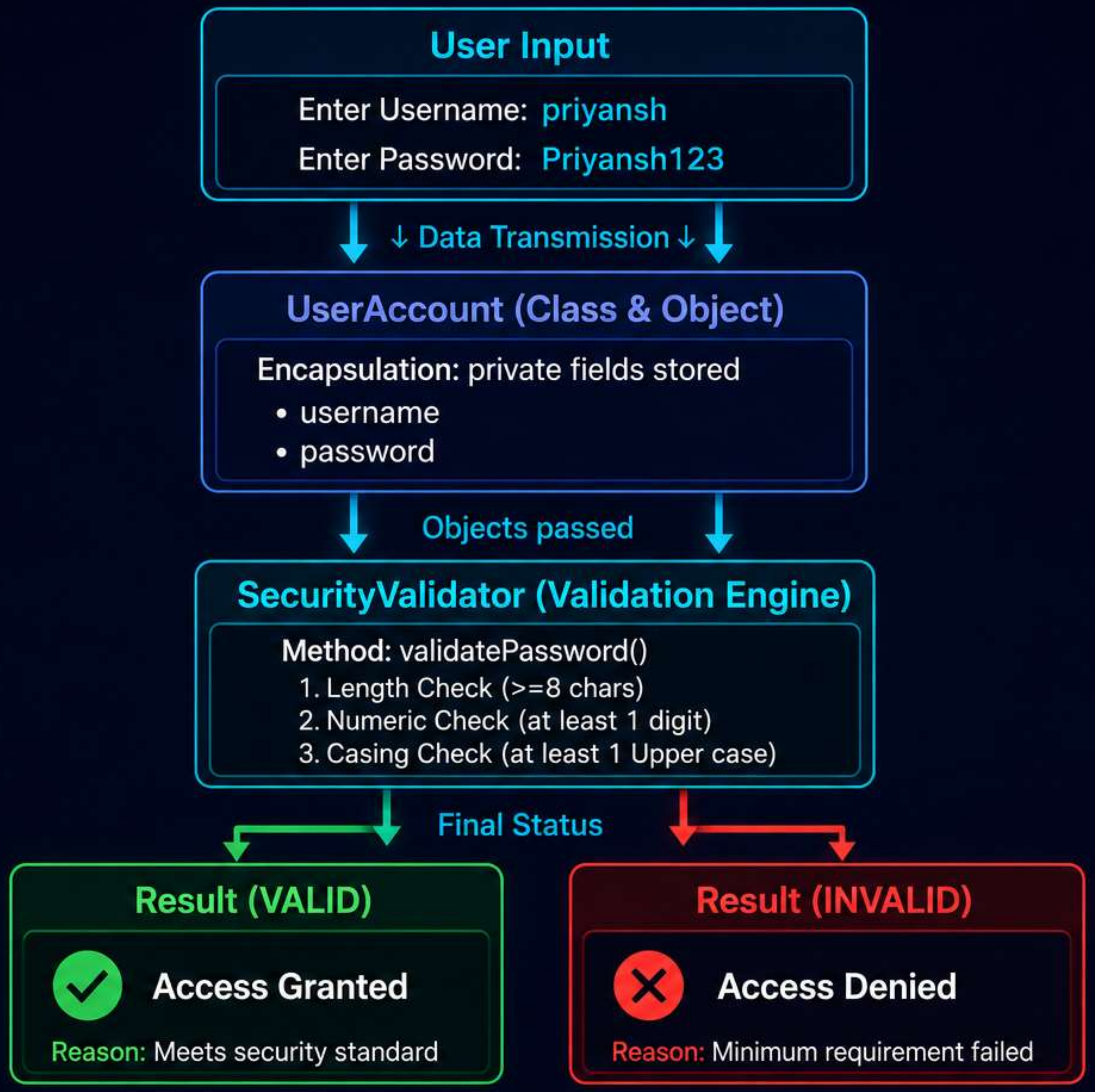
Init
UserAccount object created

Processing
Logic check begins

Verification
Applying security rules

Result
Final Status display

A simple yet effective flow ensures data travels securely from input to the validator.



Flow of Password Validation based on OOP principles.

Validation Rules

Constraint	Technical Requirement	Priority
Length Check	Minimum 8 Characters	High
Numeric Check	At least 1 Digit (0-9)	Critical
Case Check	At least 1 Uppercase Letter	Medium

```
if (pass.length() ≥ 8 && hasDigit && hasUpperCase) {
    return "VALID";
}
```

Sample Execution

Case 1: SUCCESS

Username: **priyansh**

Password: **Priyansh123**

RESULT: VALID

Case 2: FAILED

Username: **priyansh**

Password: **priyansh123**

RESULT: INVALID

(Failed: Missing Uppercase)

OOP Concepts Applied

Encapsulation

Password is stored as a **private** field. This prevents external tampering and ensures that data is only accessible via authorized methods.

Modularity

By separating the **SecurityValidator** from the **UserAccount**, we achieve code that is easy to update and test independently.

Thank You!

The project successfully validates password strength using clean Java code and OOP principles, providing a foundation for modern user security.

Special Char Logic

Strength Meter

DB Integration